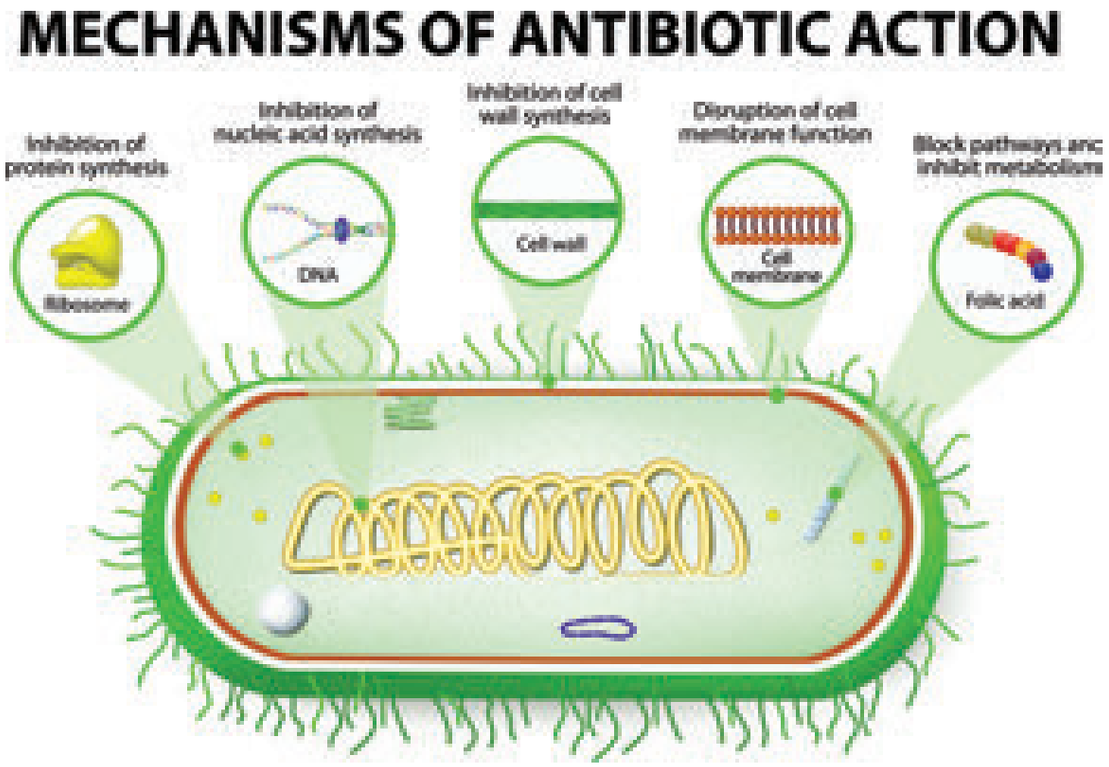


Figure 3.2: Mechanism of Antibiotic action

of pathogens refers to the limitation of pathogens to grow in the presence of effective antibiotics. There are two methods that can be used to determine the susceptibility of a potential pathogen to antimicrobial agents. They are:

- Disk diffusion method
- Tube dilution method

Disc Diffusion Method (Kirby – Bauer Test)

William Kirby and Alfred Bauer, in 1966 first introduced the principle of measuring zones of inhibition around antibiotic discs to determine antimicrobial agent susceptibilities. It is a rapid, convenient method to determine the susceptibilities of microorganisms to antimicrobial agents and a most common procedure used in susceptibility testing in clinical laboratory.

Filter paper discs containing known concentrations of antimicrobial agents are placed onto the surface of an agar plate (Muller – Hinton agar medium) inoculated with the test bacterium (Figure 3.3). The plate is incubated for 16 to 18 hours, and the zones of inhibition are read around each paper disc. During the incubation periods, the antimicrobial agent diffuses through the agar, and a concentration gradient of agent is established. (Figure) At some point in this gradient, growth of the susceptible bacteria is suppressed, and no growth is observed within a circular zone around disc. The size of a zone of inhibition must be compared to a standard Table for that particular drug before accurate comparisons can be made. Thus, enabling to classify pathogens as susceptible (S), intermediate or resistant (R) to a drug. The procedure is highly regulated and controlled by the clinical

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